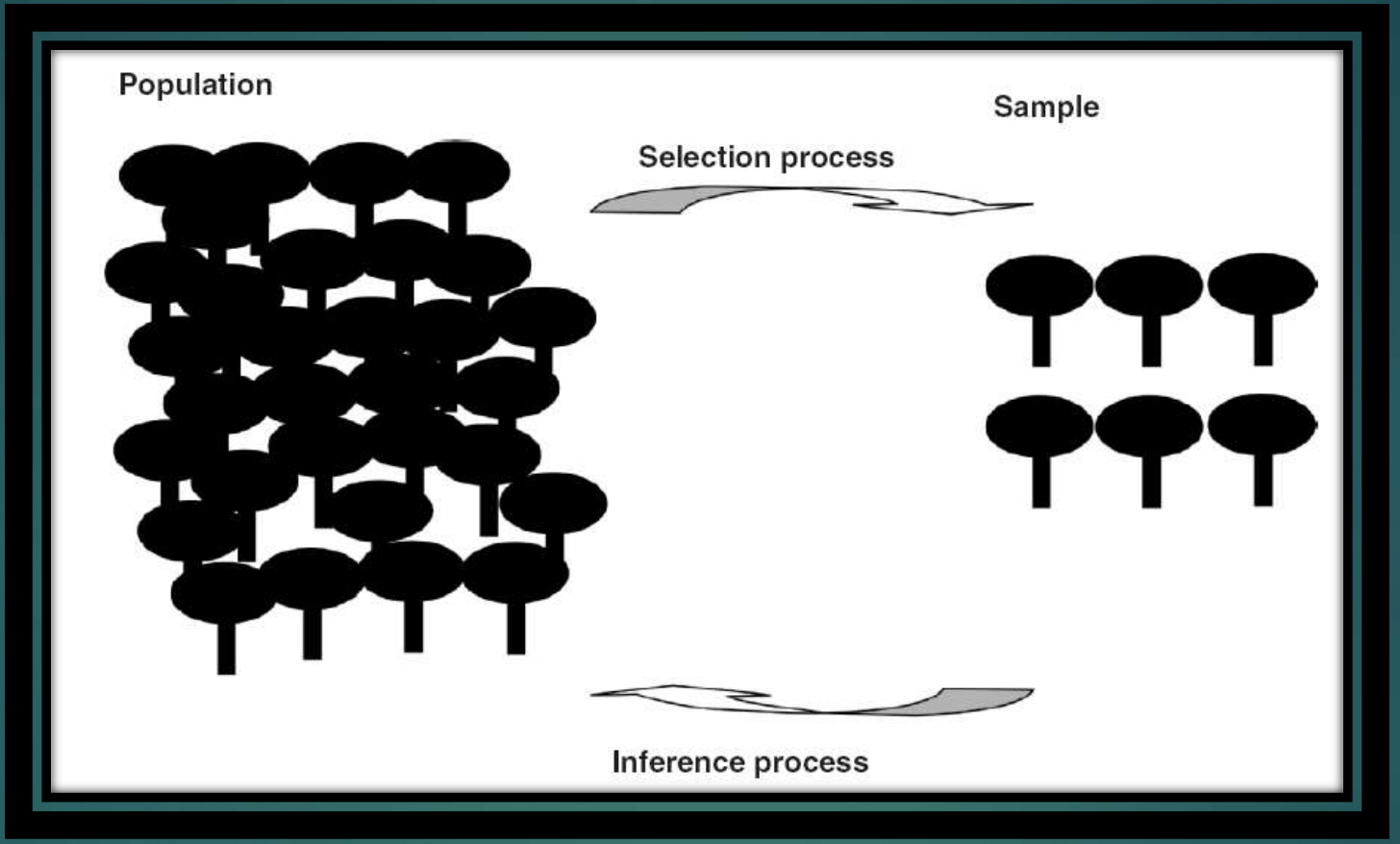
روش‌های پیشرفته نمونه برداري در جنگل

مفاهیم نمونه‌برداري جنگل

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نمونه برداري از جنگل



A population comprises all elements from which the sample is to be taken

A sample consists of a number of sampling units (or simply units) selected from the population by some design.

Parameters of forest

Averages

Aggregates

Proportions

Ratios

Each sample unit and population element possess a series of attributes of interest.

The attribute may be intrinsic or derived.

The cellulose content would be an example of an intrinsic property.

A market value, is an example of a derived attribute – an attribute that can only be obtained via another attribute or several other attributes.

Averages

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$s^2 = \sum \frac{(x_i - \bar{x})^2}{n-1}$$

$$S = \sqrt{S^2}$$

$$SE = \frac{s}{\sqrt{n}}$$

$$E = \pm t_{n-1,\frac{\alpha}{2}} SE$$

$$CI = \bar{x} \pm E$$

Totals

$$Total_x = \frac{N}{n}\sum x_i = N \times \bar{x}$$

$$SE = N \times S_{\bar{x}}$$

$$E = t_{n-1,\frac{\alpha}{2}} SE$$

$$CI_{Total} = Total \mp E$$

Proportions

Binomial

$$p = \frac{\sum x_i}{n} = \frac{e}{n}$$

$$s^2 = p(1 - p)$$

$$s = \sqrt{s^2}$$

$$s_p = \frac{s}{\sqrt{n}}$$

$$E = \pm z_{\frac{\alpha}{2}} s_p$$

$$CI = \bar{x} \pm E$$

Proportions

If more than one observation is made on the binary trait in every sampled unit the estimation of proportions has to be modified since the number of observations per sample unit can vary.
Fixed-area plots are a typical example.
The number of, say, trees per plot varies naturally between plots, The estimators of P and the variance of P are now given by

$$p = \frac{\sum_{i=1}^{n}\sum_{j=1}^{m} x_i}{\sum_{j=1}^{m} m_j} = \frac{\sum_{j=1}^{m} x_i}{\sum_{j=1}^{m} m_j} = \frac{e}{n}$$

Multi-observation

$$s_p^2 = \frac{1}{\bar{m}^2} \times \frac{\sum e_i^2 - 2p\sum e_i m_i + p^2 \sum m_i^2}{n-1}$$

Ratios

$$\hat{R} = \frac{\sum x_i}{\sum y_i} = \frac{\bar{x}}{\bar{y}}$$

$\hat{R}$ *is a biased estimator of* R

$$\text{var}_{\hat{R}} = \frac{\text{var}(y) + \hat{R} \times \text{var}(x) - 2\hat{R} \times \text{cov}(x, y)}{n \times \bar{x}^2}$$

$$\text{cov}(x, y) = \frac{\sum x_i y_i - n^{-1} \sum x_i \sum y_i}{n - 1}$$

اهداف نمونه برداري

Estimation

Estimation is quantitative inference on a fixed target quantity

Hypothesis testing

When decisions are based on an estimate of the global mean, the areal fraction or a percentile (e.g., median or P95), one must take account of the uncertainty in the estimate.
A statistical tool for decision making in the presence of uncertainty is statistical testing of hypotheses.

Classification

The term classification is used here for assigning an object to a class under uncertainty about the properties of the object.

Detection

The question is whether there is infection at all in the area, regardless where.

نمونه برداري از جنگل

- Continuum Population
- Discrete Population

- Per tree sampling
- per area sampling

- Per tree estimation
- per area estimation

رويكردهاي نمونه برداري (Sampling Approaches)

- نمونه برداري در مكان (Sampling in space)
- نمونه برداري در زمان (Sampling in time)
- نمونه برداري در مكان و زمان (Sampling in space-time)

سبك نمونه برداري (Sampling Mode)

Modes of sampling unit selection

- Convenience Sampling
- Purposive sampling
- Probability sampling

Convenience Sampling

The concept of convenience sampling is self-explanatory.

An obvious example is when sampling is limited to roadsides or other easily accessible spots.

The advantage of this mode is that it saves time and costs.

The disadvantage is that the statistical properties are inferior to those of the other modes.

For instance, estimates from a convenience sample have to be regarded as biased unless one is willing to accept specific assumptions about the sampling process and the spatial and temporal variation.

Purposive sampling

Purposive sampling tries to select the sampling units such that a given purpose is served best.

An example is the ‘free survey’ method of mapping soil classes, whereby the surveyor selects the sampling locations that are expected to be most informative with respect to soil class delineation.

In this example, the locations are selected in a subjective manner, using experience, visible landscape features and pedogenetic hypotheses, such that the surveyor expects the most useful information from his observations.

Probability sampling

Probability sampling, unlike the other modes, selects sampling units at random.

If this is done properly (according to a well-defined sampling design) the probabilities of selecting the units are known, and these probabilities provide the basis for statistical inference from the data.

مدل مفهومي نمونه برداري

- Protocol
- طراحي نمونه برداري Sampling design
  - روش نمونه برداري Sampling Method
    - نمونه برداري نقطه‌اي Point Sampling
    - نمونه برداري فاصله‌اي Distance Sampling
    - نمونه برداري خطي Line Sampling
    - نمونه برداري قطعه‌اي Plot Sampling
  - الگوي نمونه برداري Sampling patterns
    - تصادفي ساده Simple Random S.
    - بلوك تصادفي Block Random S.
    - سيستماتيک تصادفی ساده Systematic S.
    - سيستماتيک مبتنی بر خطوط ميزان (Contour sampling)
  - اجتماع نمونه Sampling clusering
    - نمونه واحد Single s.
    - نمونه خوشه‌اي Cluster S.
  - استراتژي نمونه برداري Sampling strategies
    - مونه بندي
      - پيشا مونه بندي(Prestratification)
      - پسا مونه بندي (Poststratification)
    - نمونه برداري چند گانه Multiphase S.
      - Two phase sampling with regression estimator
      - Two phase sampling with ratio estimator
      - Two phase sampling for stratification
      - .....
    - نمونه برداري چند گامي Multistage S.
      - Two stage sampling
      - Three stage sampling
      - .....

کدام روش ؟

- از نظر آماري (Statistically)
- The most precise and accurate
- از نظر تخصصي (Specialty)
- the easiest to set and use
- از نظر منطقي (Logically)
- The most efficient to answer the question

عوامل موثر بر انتخاب روش

- بر اساس اهداف آماربرداري
  - متغيرهاي مورد بررسي
- بر اساس ويژگي‌هاي جنگل
  - تراكم
  - الگوي پراكنش مكاني

نحوه انتخاب نمونه یا فرد در نمونه

- انتخاب به نسبت فراواني (PPF)
- انتخاب به نسبت اندازه (PPS)

منابع این جلسه

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Gruijter **et al. 2006**
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Chapter 1: Introduction
Chapter 2: Modes of sampling and statistical inference

Sampling Methods, Remote Sensing And GIS Multiresource Forest Inventory
Kohl et al**. 2006**
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Chapter 3: Sampling in forest survey